Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 47

Solving Linear Recurrences

Covers *MCS* Section 21.3

Status of this set

Ungraded practice. Solutions released with it. **Check your closed form against the first few terms** — it costs one line and catches almost every error.

Part I: How to Practice This Material

Homogeneous: three steps

1. Write the characteristic equation and find its roots.
2. Each simple root r gives a term cr^n; a root of multiplicity k gives $r^n, nr^n, \ldots, n^{k-1}r^n$.
3. Apply the boundary conditions to solve for the constants.

Inhomogeneous: add two steps

Between steps 2 and 3, find a *particular* solution by guessing a form that matches $g(n)$ — a constant for a constant, a polynomial of the same degree for a polynomial, cr^n for an exponential. Add it to the homogeneous solution, *then* apply the boundary conditions.

Three mistakes to expect

1. **Applying the boundary conditions before adding the particular solution.** They must be applied to the *general* solution, last.
2. **Using two copies of r^n for a repeated root.** Use r^n and nr^n.
3. **Sign errors reading off the characteristic equation.** $f(n) = 5f(n-1) - 6f(n-2)$ gives $x^2 - 5x + 6 = 0$.

Part II: Practice Problems

Section A — Distinct roots

1. Solve $f(n) = f(n-1) + 6f(n-2)$ with $f(0) = 1$ and $f(1) = 8$.

Section B — Repeated roots

2. Solve $f(n) = 6f(n-1) - 9f(n-2)$ with $f(0) = 1$ and $f(1) = 9$.

Section C — Inhomogeneous

3. Solve $f(n) = 2f(n-1) + 3$ with $f(0) = 1$.

4. Solve $f(n) = 3f(n-1) + 2$ with $f(0) = 0$.

Section D — Writing

5. In five to eight complete sentences, explain why guessing $f(n) = x^n$ is the right first move for a homogeneous linear recurrence, and what the characteristic equation is actually telling you.